

Ann Song

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EDUCATION

Yale University

BS in Computer Science and Mathematics | GPA: 3.93/4.0

New Haven, CT

June 2029

TA for CPSC 2020 Mathematics for Computer Science, Freshmen Pre-Orientation Leader, Summer 2026 Tour Guide

WORK EXPERIENCE

Medtronic (Surgical Robotics Operating Unit)

North Haven, CT

Software Engineer Intern

June 2026 – Aug 2026

- Worked in the Signia powered surgical stapler's embedded C firmware codebase running Micrium μ C/OS-II, adding persistent diagnostics for scheduling jitter, task utilization, reset causes, and system faults on the motor-control path to improve post-failure observability of a high-reliability medical device.
- Redesignated a legacy device-log analysis pipeline as a modular Python system, achieving $3\times$ faster processing on full device logs while expanding support to 12 record types across 4 firmware versions used for returned-device failure investigation.
- Built automated telemetry analysis that reconstructs device state and motor behavior across clamp, staple, and cut phases, enabling engineers to isolate failures to specific software or electromechanical subsystems from raw device captures.
- Interviewed 14 Release Product Engineering, Systems, and R&D stakeholders to identify gaps in existing diagnostic tooling, then built a unified analysis interface that reduced returned-device triage time by 60%.

AI Engineer (Contract)

Oct 2025 – May 2026

Software Engineer Intern

June 2025 – Aug 2025

- Designed and shipped a production Python RAG system for Software Failure Modes and Effects Analysis (SFMEA), translating requirements from Systems, Quality, and field teams into software used by 800+ surgical stapler sales representatives and reducing engineering escalations by 65%.
- Architected the retrieval layer over 10,000+ pages of specifications and device-architecture documentation using semantic chunking and document classification, deployed as Python microservices through a CI/CD pipeline.
- Built an automated evaluation framework comparing 6 LLMs on latency, cost, accuracy, and hallucination rate against representative surgeon questions, reducing inference cost by 60% while maintaining response accuracy.

Yale Inquisitive Robotics Lab

New Haven, CT

Undergraduate Researcher in Physical AI

Sept 2025 – Present

- Built data and evaluation pipelines for failure prediction in Vision-Language-Action policies, processing internal model representations and multimodal behavior across 3,000+ robot manipulation rollouts from 3 VLA models; second-author manuscript in submission to ICLR 2027.
- Evaluated 40+ diagnostic signals and trained sequential models over 512-dimensional representations with rollout-level train/validation/test isolation and evaluation across 20 randomized seeds to measure robustness and generalization.

PROJECTS

Autonomous Tomato Harvesting Robot (2-Person Team) [Robot Demo](#)

New Haven, CT

ROS 2, C++, Python, OpenCV, YOLOv11, Linux, Gazebo, Raspberry Pi

June 2026 – Present

- Built and deployed an autonomous cherry-tomato harvesting system on a custom 4-DOF robotic arm, integrating perception, stereo localization, motion planning, motor control, and end effector in ROS 2 on Linux.
- Implemented a closed-form 4-DOF inverse-kinematics solver and staged pregrasp–contact–retreat planner that convert 3D targets into collision-conscious, reachable joint goals while maintaining the end-effector orientation required for suction grasping.
- Developed a feedback-driven motion controller that generates coordinated joint trajectories and scales per-joint velocity by angular travel so all joints arrive simultaneously, producing repeatable straight-line approaches on the physical arm.
- Built a calibrated stereo-vision pipeline using dual Raspberry Pi cameras, rectification, semi-global block matching, and camera-to-robot transforms to estimate tomato position in the robot frame for closed-loop manipulation.
- Deployed YOLOv11 detection with multi-target tracking and independent color-based ripeness validation, prioritizing precision and rejecting uncertain or unripe targets before motion commands are issued.
- Built a Gazebo test environment from the robot URDF and used ground-truth target poses to isolate perception failures from kinematics and control failures, enabling software debugging before and alongside testing on physical hardware.

SKILLS AND INTERESTS

Languages: Python [NumPy, SciPy, PyTorch, Pandas, OpenCV], C++, C, Java, JS [ReactJS, TypeScript], MATLAB, Bash

Robotics: ROS 2, URDF, TF2, RViz, Gazebo, Foxglove, stereo calibration, SGBM, forward/inverse kinematics, YOLO

Tools: Git, Docker, Linux, AWS, JAX/Flax, Slurm/HPC, Claude Code, Cursor

Interests: Yale Red Lantern (Traditional Chinese Dance), Yale Figure Skating Club, Weightlifting